# From Stencils to Group Rings

**A ground-up primer on $N_c = \dim \ker A$, fractal subsystem symmetries, and the closed form for the `haah_z` stencil**

Compiled notes — Zettelkasten `500_masters`

September 1, 2026

**Abstract**

This primer builds, from the definition of a group, every piece of algebra needed to understand one sentence: *rolling a local stencil across a periodic lattice is multiplication in a group ring, and the ground-state degeneracy is the dimension of the kernel of that multiplication map.* It then uses that machinery to derive, explain and verify a closed form for $N_c(L_x, L_y)$ for the two-dimensional Haah-type corner stencil $f = 1 + x + x^2 + y + xy + y^2$ over GF(2), and closes with a reading of the sweep code and of what the kernel vectors mean physically: they are the model's **fractal subsystem symmetries**, and counting them is counting the symmetries. Every numerical claim in the text has been re-derived independently and is reported in §18. Nothing here assumes more than linear algebra.

## Contents

**Part I — The vocabulary** **3**

**1 The whole story in one page** **3**

**2 Groups: a set with one operation** **4**

**3 Rings and fields: the arithmetic your bits already use** **4**

**4 Polynomial rings: a pattern as a single object** **4**

**5 Quotient rings: teaching algebra to wrap around** **5**

**6 Group rings: the meeting point** **5**

**7 Circulant matrices: what the code actually builds** **6**

**Part II — What the kernel means** **6**

**8 Ground states, symmetries, and why they are the same set** **6**

**9 Translation operators, momentum sectors, and one tempting wrong guess** **7**

**10 Worked example: the toric code, where the answer is size-independent** **8**
10.1 The analytic shortcut: cyclic-code theory . . . . . 9

**11 Doing it without the polynomial** **9**

**12 Fractal subsystem symmetries and UV/IR mixing** **11**

**Part III — The closed form** **12**

**13 Statement** **12**

**14 Derivation** **13**
14.1 Splitting the ring . . . . . 13
14.2 The curve has exactly one singular point . . . . . 13
14.3 Local multiplicities . . . . . 14
14.4 The conic factors over $\mathbb{F}_4$: the fractal stencil is two straight lines . . . . . 14

**15 Explicit special cases** **14**

**16 When is $N > 1$? The number-theoretic part** **16**

**17 The method generalises: Newman–Moore in closed form** **16**

**18 Numerical validation** **17**

**Part IV — Reading the code** **18**

**19 `nc_sweep.py`, line by line** **18**

**20 Scope and open directions** **19**

**A Glossary** **19**

**B Reproducing the numbers** **20**

# Part I — The vocabulary

## 1 The whole story in one page

Before any algebra, here is the entire object of study, stated concretely.

Put one variable $t_{\vec{r}} \in \{0,1\}$ on every site $\vec{r}$ of an $L_y \times L_x$ grid, and make the grid periodic in both directions (a torus). Fix a small fixed *shape* — a **stencil** — a list of offsets $\vec{s}$ with coefficients $c_{\vec{s}}$. Stamp that shape down at every site, and demand that the sum of the variables under the stamp is zero mod 2:

$$(At)_{\vec{r}} \;=\; \sum_{\vec{s}} c_{\vec{s}}\, t_{\vec{r}+\vec{s}} \;\equiv\; 0 \pmod 2 \qquad \text{for every } \vec{r}. \tag{1}$$

That is $L_xL_y$ linear equations in $L_xL_y$ unknowns over the two-element field. The solutions form a vector space, and

$$\boxed{N_c \;=\; \dim_{\mathrm{GF}(2)} \ker A \;=\; L_xL_y - \operatorname{rank} A, \qquad \mathrm{GSD} = 2^{N_c}.} \tag{2}$$

Everything in this document is commentary on (2). One caveat belongs here rather than in a footnote: (2) in that form is the *single-species, one-variable-per-site* case. The general statement carries an extra term and is derived in §11; the polynomial machinery of Part I is a tool for evaluating it, not a prerequisite for stating it. There are three questions:

(A) **What is $\ker A$, physically?** It is simultaneously the ground-state space of a classical multi-spin Hamiltonian *and* the group of spin-flip symmetries of that Hamiltonian. Those two things coincide for a reason (§8), and the symmetries turn out to be supported on *fractals* (§12).

(B) **Can $N_c$ be computed without Gaussian elimination?** Yes — $A$ is multiplication by a polynomial in a ring, and ring theory answers the question. For the toric code the answer is a one-line argument giving $\mathrm{GSD} = 4$ for every lattice size (§10). For the Haah stencil the answer is Theorem 13.1 (§13).

(C) **Why does $N_c$ depend on $L_x, L_y$ so erratically?** Because the answer is a point count on a curve over a finite field, and those counts are number-theoretic (§16). That erratic dependence *is* UV/IR mixing, and it is the fingerprint that separates a fracton-like phase from ordinary topological order.

**The dictionary, stated once**

$$\begin{aligned}
\text{a site at offset } (d_1,d_2) &\longleftrightarrow \text{the monomial } x^{d_1}y^{d_2}\\
\text{a stencil} &\longleftrightarrow \text{a polynomial } f(x,y)\\
\text{a spin configuration} &\longleftrightarrow \text{a polynomial } t(x,y)\\
\text{stamping and summing (convolution)} &\longleftrightarrow \text{multiplication } f\cdot t\\
\text{periodic boundary conditions} &\longleftrightarrow \text{the relations } x^{L_x}=1,\ y^{L_y}=1\\
\text{the matrix the code builds} &\longleftrightarrow \text{“multiply by } f\text{” in } \mathbb{F}_2[x,y]/(x^{L_x}-1, y^{L_y}-1)\\
N_c &\longleftrightarrow \dim\ker(\text{that map}) \;=\; \dim \mathrm{Ann}(f)
\end{aligned}$$

The rest of Part I earns each line of that table.

## 2 Groups: a set with one operation

**Definition 2.1** (Group)**.** A *group* is a set $G$ with one operation (written $+$ or $\cdot$) such that (i) combining two elements stays in $G$ (closure); (ii) bracketing does not matter (associativity); (iii) there is a neutral element $e$; (iv) every $g$ has an inverse $g^{-1}$ with $g \cdot g^{-1} = e$.

**Example 2.2** (The integers)**.** $\mathbb{Z}$ under addition: identity 0, inverse $-n$.

**Example 2.3** (The one that matters)**.** Fix $L$. The set $\mathbb{Z}_L = \{0, 1, \ldots, L-1\}$ with addition *modulo* $L$ is a group: shifting by $a$ then by $b$ is shifting by $a + b \bmod L$. This is exactly "positions on a ring of $L$ sites". A two-dimensional periodic lattice is $G = \mathbb{Z}_{L_x} \times \mathbb{Z}_{L_y}$: pairs of positions, each wrapping independently.

**What "periodic boundary conditions" really says**

The set of sites of your lattice, *together with translation*, is a finite abelian group. Nothing more is meant by PBC. Every subsequent construction is built on top of this one group.

Note what a group cannot do: it has one operation. To write $1 + x + y^2$ we need two — one to combine terms, one to combine positions. That is a ring.

## 3 Rings and fields: the arithmetic your bits already use

**Definition 3.1** (Ring)**.** A *ring* is a set $R$ with two operations $+$ and $\times$ such that $(R, +)$ is a commutative group with identity 0; $\times$ is associative and distributes over $+$; and there is a multiplicative identity 1. We only ever need *commutative* rings. A *field* is a commutative ring in which every non-zero element has a multiplicative inverse.

**Example 3.2** ($\mathbb{F}_2 = \mathrm{GF}(2)$)**.** $\{0, 1\}$ with addition mod 2 (XOR) and ordinary multiplication. It is a field: 1 is its own inverse. This is the arithmetic your $t_{\vec{r}} \in \{0, 1\}$ already live in — "ring" is simply the general name for the kind of system $\mathbb{F}_2$ is one example of.

**Example 3.3** ($\mathbb{F}_4$, needed in Part III)**.** $\mathbb{F}_4 = \{0, 1, \omega, \omega^2\}$ where $\omega^2 + \omega + 1 = 0$ and $\omega^3 = 1$. Concretely $\mathbb{F}_4 = \mathbb{F}_2[\omega]/(\omega^2 + \omega + 1)$: elements are $a + b\omega$ with $a, b \in \mathbb{F}_2$, added bitwise and multiplied using $\omega^2 = \omega + 1$. It contains $\mathbb{F}_2$, and squaring $u \mapsto u^2$ (the *Frobenius* map) swaps $\omega \leftrightarrow \omega^2$ while fixing $0, 1$. Two facts we will use constantly: in characteristic 2, $(a+b)^2 = a^2 + b^2$; and $\mathbb{F}_4 \subset \mathbb{F}_{2^d}$ if and only if $d$ is even.

*Remark* 3.4 (Why characteristic 2 is not a technicality). In char 2, $+$ and $-$ are the same thing, squaring is a ring homomorphism, and $x^L - 1 = (x^m - 1)^{2^a}$ whenever $L = 2^a m$. That last identity is the entire reason the final answer splits into "a power-of-two part" and "an odd part" (§14.1).

## 4 Polynomial rings: a pattern as a single object

Take coefficients in $\mathbb{F}_2$ and let $x$ be a formal symbol.

**Definition 4.1.** $\mathbb{F}_2[x]$ is the set of expressions $a_0 + a_1 x + \cdots + a_n x^n$ with $a_i \in \mathbb{F}_2$, added coefficientwise mod 2 and multiplied by expanding and collecting like terms. $\mathbb{F}_2[x, y]$ is the same with two symbols.

**The reinterpretation that makes everything work**

Read $x^a y^b$ **not** as "a variable raised to a power" but as **"the lattice site at position** $(a, b)$**"**. Then a polynomial $t(x, y) = \sum_{a,b} t_{ab} x^a y^b$ is just a compact name for a 0/1 pattern on the lattice: it is 1 at the sites where $t_{ab} = 1$.

**Example 4.2.** An operator touching the origin, the site one step right, and the site two steps up is $f(x,y) = 1 + x + y^2$. The Haah-type stencil of this document,

$$\texttt{HAAH_Z_SHIFTS} = \{(0,0),(0,1),(0,2),(1,0),(1,1),(2,0)\} \quad \text{(rows, cols)},$$

is the triangular corner shape

$$f(x,y) \;=\; 1 + x + x^2 + y + xy + y^2. \tag{3}$$

**Proposition 4.3** (Multiplication *is* convolution)**.** *If* $f = \sum_a f_a x^a$ *and* $t = \sum_b t_b x^b$ *then* $ft = \sum_c \big(\sum_a f_a t_{c-a}\big) x^c \pmod 2$.

The coefficient of $x^c$ is obtained by sliding $t$'s pattern until it lines up at offset $c$, multiplying overlaps and summing mod 2. That is precisely "roll the stencil across the lattice and combine". **Rolling a stencil and multiplying polynomials are the same operation written in two notations.** This is the single most important sentence in Part I.

## 5 Quotient rings: teaching algebra to wrap around

$\mathbb{F}_2[x]$ describes an infinite lattice — there is no largest power of $x$. Your lattice is finite and periodic. We must tell the algebra that.

**Definition 5.1** (Quotient ring)**.** Given a ring $R$ and $m \in R$, the quotient $R/(m)$ is $R$ with the extra rule that $m$ is declared equal to 0 (hence so is any multiple of $m$). Two elements are "the same" in $R/(m)$ exactly when their difference is a multiple of $m$.

**Example 5.2** (You already know this)**.** $\mathbb{Z}/(12)$ is clock arithmetic: $12 = 0$, so 13 and 1 are the same element.

**Example 5.3** (The one you need)**.** $\mathbb{F}_2[x]/(x^L - 1)$ is $\mathbb{F}_2[x]$ with the extra rule $x^L = 1$. Since $x^a$ means "site $a$", this says: site $L$ is site 0, site $L+1$ is site 1, and so on. *This is periodic boundary conditions written algebraically.* Its elements are described by $x^0, \ldots, x^{L-1}$ — exactly the $L$ independent sites. In two dimensions,

$$\mathcal{R}_L \;=\; \mathbb{F}_2[x,y]/\big(x^{L_x} - 1,\; y^{L_y} - 1\big), \qquad \dim_{\mathbb{F}_2} \mathcal{R}_L = L_x L_y.$$

## 6 Group rings: the meeting point

**Definition 6.1** (Group ring)**.** Given a group $G$ and a coefficient ring $R$, the *group ring* $R[G]$ is the set of formal sums $\sum_{g \in G} c_g\, g$ with $c_g \in R$, added coefficientwise and multiplied by using $G$'s own operation on the labels:

$$\Big(\sum_g a_g g\Big)\Big(\sum_h b_h h\Big) \;=\; \sum_k \Big(\sum_{gh=k} a_g b_h\Big) k.$$

Take $G = \mathbb{Z}_{L_x} \times \mathbb{Z}_{L_y}$ and $R = \mathbb{F}_2$, and write the group element $(a,b)$ as the symbol $x^a y^b$. The definition above becomes, verbatim, the quotient ring of the previous section:

$$\boxed{\mathbb{F}_2\big[\mathbb{Z}_{L_x} \times \mathbb{Z}_{L_y}\big] \;\cong\; \mathbb{F}_2[x,y]/\big(x^{L_x} - 1,\; y^{L_y} - 1\big) \;=\; \mathcal{R}_L.} \tag{4}$$

**Why the two names describe one object**

A group ring built on a group of *translations* is automatically a ring of *periodic patterns*, and its multiplication is automatically *convolution* — one pattern rolled across another, wrapping at the boundary. That is exactly what your stencil does. The abstraction adds no assumption; it renames an operation you already perform, so that theorems about rings become usable.

## 7 Circulant matrices: what the code actually builds

Multiplication by $f$ is an $\mathbb{F}_2$-linear map $\mathcal{R}_L \to \mathcal{R}_L$. Choose the basis $\{x^a y^b\}$ and you get a matrix $A$ of size $L_x L_y$. Because multiplication by $x$ is a cyclic shift, every column of $A$ is a shifted copy of the previous one: $A$ is *circulant* in one dimension, *block-circulant* in two.

**Example 7.1** (The miniature example, done fully by hand)**.** Take $L = 4$ in one dimension and $f(x) = 1 + x$. Acting on $t = t_0 + t_1 x + t_2 x^2 + t_3 x^3$ in $\mathbb{F}_2[x]/(x^4 - 1)$:

$$A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \pmod 2,$$

where column $j$ records $f(x) \cdot x^j$. Row-reducing gives $\operatorname{rank} A = 3$, so $\dim \ker A = 1$, spanned by $t = (1, 1, 1, 1)$: indeed $(1 + x)(1 + x + x^2 + x^3) = 1 + x^4 \equiv 1 + 1 = 0$. A two-site stencil became a $4 \times 4$ circulant matrix purely mechanically, and finding the kernel became ordinary Gaussian elimination. That is everything `nc_sweep.py` does, at larger scale.

**Proposition 7.2** (Rank–nullity, over any field)**.** *For $A : V \to V$ with $\dim V = N$, $\dim \ker A = N - \operatorname{rank} A$. This holds over $\mathrm{GF}(p)$ exactly as over $\mathbb{R}$.*

So $N_c$ is not a new quantity requiring new machinery. It is one null-space computation. Equivalently, in ring language,

$$\ker A \;=\; \mathrm{Ann}_{\mathcal{R}_L}(f) \;=\; \{\, t \in \mathcal{R}_L : f \cdot t = 0 \,\}, \tag{5}$$

the *annihilator ideal* of $f$. Everything analytic that follows is a computation of an annihilator.

# Part II — What the kernel means

## 8 Ground states, symmetries, and why they are the same set

The stencil defines a classical Hamiltonian with one multi-spin term per site. Writing $\sigma_{\vec{r}} = (-1)^{t_{\vec{r}}}$ and taking all $c_{\vec{s}} = 1$,

$$\mathcal{H} \;=\; -J \sum_{\vec{r}} \prod_{\vec{s} \in S} \sigma_{\vec{r}+\vec{s}}. \tag{6}$$

A configuration is a ground state iff every product is $+1$, i.e. iff $(At)_{\vec{r}} = 0$ for all $\vec{r}$: the ground-state space is $\ker A$, and $\mathrm{GSD} = 2^{N_c}$.

Now ask a different question: which *spin flips* are symmetries? Flip the spins on the support of a pattern $u$, i.e. $\sigma_{\vec{r}} \mapsto (-1)^{u_{\vec{r}}} \sigma_{\vec{r}}$. The term at $\vec{r}$ picks up the factor $(-1)^{\sum_{\vec{s}} u_{\vec{r}+\vec{s}}}$, so *every* term is left invariant precisely when

$$\sum_{\vec{s} \in S} u_{\vec{r}+\vec{s}} = 0 \quad \forall \vec{r} \qquad \Longleftrightarrow \qquad u \in \ker A^{\top}, \tag{7}$$

where $A^{\top}$ is the operator of the *reversed* stencil $\vec{s} \mapsto -\vec{s}$ (this is the relation between `haah_z` and `haah_x` in the code). Since $\operatorname{rank} A = \operatorname{rank} A^{\top}$, the two kernels have the same dimension.

**The identification, and why it is not a coincidence**

For a *linear* stencil model the ground-state space and the group of flip symmetries are both $N_c$-dimensional $\mathbb{F}_2$-vector spaces, and the symmetry group acts simply transitively on the ground states. Concretely: fix any ground state $t^{(0)}$; then every other ground state is $t^{(0)} + u$ with $u$ a symmetry, and conversely. Counting ground states and counting symmetries are literally the same count. Hence

$$\text{GSD} \;=\; \big|\{\text{flip symmetries}\}\big| \;=\; 2^{N_c}.$$

*Remark* 8.1 (This settles an open to-do). The vault note *Ground State Degenerancy Counter* asks whether the kernel basis vectors "really are symmetries of the original Hamiltonian". They are, by the two-line computation above — with one caveat worth stating precisely: the flip symmetries are $\ker A^{\top}$, not $\ker A$. For the Haah stencil the two are the `haah_x` and `haah_z` kernels, which the sweep confirms are equidimensional for every lattice; for a *symmetric* stencil (e.g. $\Delta_6$) they coincide as sets.

*Remark* 8.2 (Two conventions in circulation). In the *stabilizer-code* reading (toric code, §10) one instead writes the Pauli stabilizers as polynomials and asks for *relations* among them — products of stabilizers equal to the identity. There ker counts *redundancies*, and GSD $= 2^{\#\text{qubits}-\#\text{independent stabilizers}}$. In the *classical stencil* reading used above, ker counts *ground states* directly. The two are the same linear algebra applied at different places; keep track of which one a formula is using. When the code carries both an $X$ and a $Z$ sector, the degeneracy is GSD $= 2^{2N_c}$, whereas `nc_sweep.py` reports the single-sector $2^{N_c}$.

## 9 Translation operators, momentum sectors, and one tempting wrong guess

It is worth seeing where the group ring comes from on the quantum side, because the naive route fails instructively.

On a chain of $N$ sites with PBC, the translation operator is $\hat{T}_a = \sum_n |n+1\rangle\langle n|$, i.e. $\hat{T}_a |n\rangle = |n+1\rangle$. It is unitary (it permutes basis states) and is diagonalised by Bloch states

$$|k\rangle = \tfrac{1}{\sqrt{N}} \sum_n e^{ikna} |n\rangle\,, \qquad \hat{T}_a |k\rangle = e^{-ika} |k\rangle\,, \qquad k \in (-\pi/a, \pi/a],$$

so $\hat{T}_a = e^{-i\hat{p}a/\hbar}$: the lattice's only difference from the continuum is that momentum is defined modulo $2\pi/a$.

If $\mathcal{H}$ is translation invariant then $\hat{T}_a$ is the unique unitary with $\hat{T}_a \sigma_n^{\alpha} \hat{T}_a^{\dagger} = \sigma_{n+1}^{\alpha}$ for every local operator appearing in $\mathcal{H}$; $[\hat{T}_a, \mathcal{H}] = 0$ then follows term by term, because relabelling $n \to n+1$ cyclically permutes the sum. Concretely $\hat{T}_a$ is the cyclic shift of tensor factors, buildable as a product of adjacent SWAPs.

Summing over a full orbit, $F = \sum_{n=0}^{N-1} \hat{T}_a^{\,n}$, and acting on a momentum eigenstate, the geometric sum gives $F|\psi_k\rangle = N|\psi_k\rangle$ if $k = 0$ and 0 otherwise. So $F = N\hat{P}_{k=0}$, and more generally $F_k = \sum_n e^{ikna} \hat{T}_a^{\,n} = N\hat{P}_k$.

**$F$ is a diagnostic, not a cause**

$F_k$ tells you which momentum sector a state lives in. If $[\hat{T}_a, \mathcal{H}] = 0$ and the ground state is unique it automatically carries definite momentum. When the ground state is *degenerate* — from SSB, a Lieb–Schultz–Mattis filling constraint, or topological order — $F_k$ is the tool that sorts the degenerate states by momentum.

**The tempting wrong guess**

One might hope to count redundancies as $\dim\ker \hat{T}_{\vec{a}}$. This is always zero: translations are unitary, so their kernel is trivial in any dimension. The kernel that counts is the kernel of the *constraint*, not of the translation. Diagonalising all commuting $\hat{T}_{\vec{a}}$ simultaneously (a lattice Fourier transform) turns the family of $N$ constraints into a single $\vec{k}$-dependent matrix $C(\vec{k})$ on the internal index, and

$$\#\{\text{independent constraints}\} = \sum_{\vec{k}\in \mathrm{BZ}} \operatorname{rank} C(\vec{k}), \qquad \#\{\text{redundancies}\} = \sum_{\vec{k}\in \mathrm{BZ}} \dim\ker C(\vec{k}).$$

Over GF(2) the "Fourier transform" is exactly the ring splitting of §14.1, with $\vec{k}$ replaced by pairs of roots of unity.

## 10 Worked example: the toric code, where the answer is size-independent

This is the cleanest end-to-end use of the machinery, and the control case against which fracton behaviour is defined.

**Setup.** Square lattice, $L_1 \times L_2$ unit cells on a torus, two qubits per cell (a horizontal edge $h$ and a vertical edge $v$):

$$A_{\vec{n}} = X_{h(\vec{n})}X_{h(\vec{n}-\hat{x})}X_{v(\vec{n})}X_{v(\vec{n}-\hat{y})}, \qquad B_{\vec{n}} = Z_{h(\vec{n})}Z_{h(\vec{n}+\hat{y})}Z_{v(\vec{n})}Z_{v(\vec{n}+\hat{x})},$$

$\mathcal{H} = -\sum_{\vec{n}} A_{\vec{n}} - \sum_{\vec{n}} B_{\vec{n}}$.

**Paulis to polynomials.** All the $A_{\vec{n}}$ commute, so track only *which* qubits each touches: a support vector over $\mathbb{F}_2$, with multiplication of commuting Paulis becoming XOR ($X^2 = \mathbb{1}$). Encode the shift by $\vec{n} = (n_1, n_2)$ as $x^{n_1}y^{n_2}$ and read off the reference operator's support:

$$a(x,y) = \big(1 + x^{-1},\ 1 + y^{-1}\big), \qquad b(x,y) = \big(1 + y,\ 1 + x\big),$$

one component per sublattice ($h$ and $v$).

**How to read off $a(x,y)$ in general**

(1) Write down the single reference operator at the origin. (2) For each sublattice component it touches, list the relative offset $(d_1, d_2)$. (3) Each offset contributes $x^{d_1}y^{d_2}$ to that component's polynomial. That is the entire recipe — $a(x,y)$ is bookkeeping for "which qubits, at which relative positions".

**Relations as a kernel.** A product $\prod_{\vec{n}} A_{\vec{n}}^{f_{\vec{n}}}$ corresponds to $f(x,y) = \sum f_{\vec{n}} x^{n_1}y^{n_2}$, and equals the identity iff $f \cdot a = (0,0)$ componentwise. So relations $= \ker(f \mapsto f \cdot a)$.

**Infinite lattice: none.** $\mathbb{F}_2[x^{\pm 1}, y^{\pm 1}]$ has no zero divisors, so $f \cdot a = 0$ with $a \neq 0$ forces $f = 0$. Every star is independent.

**On the torus: the ring changes.** Now the ring is $\mathcal{R} = \mathbb{F}_2[x,y]/(x^{L_1}-1, y^{L_2}-1)$ — the group ring of (4) — and it *does* have zero divisors. Take $f = \sum_{i,j} x^i y^j$ ("multiply all the stars"). Shifting relabels the same finite sum, so $f \cdot x^{-1} = f$ and hence $f \cdot (1 + x^{-1}) = f + f = 0$; likewise for $y$. This $f$ is a genuine kernel element: it is the relation $\prod_{\vec{n}} A_{\vec{n}} = \mathbb{1}$ (every edge belongs to two vertices, so every $X$ cancels). Same for $b$.

**Degeneracy.** With $N = 2L_1L_2$ qubits and $L_1L_2$ stars + $L_1L_2$ plaquettes minus one redundancy each,

$$\text{rank} = 2L_1L_2 - 2, \qquad \dim(\text{GS space}) = 2^{N-\text{rank}} = 2^2 = 4,$$

the four topological sectors on the torus.

### 10.1 The analytic shortcut: cyclic-code theory

Checking lattice by lattice is unnecessary. The stabilizer ring here is the ring used to build cyclic codes, and cyclic-code theory gives a size-independent argument.

**The key fact**

In $R_L = \mathbb{F}_2[x]/(x^L - 1)$, if $g(x)$ divides $x^L - 1$, write $x^L - 1 = g(x)h(x)$. Then

$$\text{Ann}(g) = \{f : fg \equiv 0 \bmod x^L - 1\} = (h), \qquad \dim \text{Ann}(g) = \deg g,$$

for *every* $L$ with $g \mid x^L - 1$ — no case-by-case checking.

Apply it: $a_h(x) = 1 + x^{-1} \sim x + 1$ (multiplying by the unit $x$ does not change the annihilator). In characteristic 2, $x = 1$ is always a root of $x^{L_1}-1$, so $(x+1)$ always divides it, for every $L_1$. Hence $\dim \text{Ann}(a_h) = \deg(x+1) = 1$ always, with generator $h(x) = 1+x+\cdots+x^{L_1-1}$ — exactly the all-ones combination found by hand. Same for $a_v(y)$.

Because $a_h$ depends only on $x$ and $a_v$ only on $y$, the ring factors as a tensor product $R_{L_1} \otimes R_{L_2}$ and a valid $f$ must lie in $\text{Ann}(a_h) \otimes \text{Ann}(a_v)$:

$$\dim \ker(\text{star constraints}) = \dim \text{Ann}(a_h) \times \dim \text{Ann}(a_v) = 1 \times 1 = 1$$

analytically, for all $L_1, L_2$ simultaneously. Same for plaquettes. Hence GSD $= 4$ always.

**When the shortcut applies**

This product argument works whenever $a(x,y)$ splits into independent single-variable factors, one per lattice direction — true for "cross"-shaped nearest-neighbour supports and most simple gauge-theory constraints.

## 11 Doing it without the polynomial

Everything so far went through $f(x,y)$. It is worth being explicit that *none of it is required to get the number*. The rolling operation and its kernel suffice on their own, for any number of species and any number of degrees of freedom per cell, with no stencil polynomial and no sublattice bookkeeping beyond an enumeration.

**The general recipe.** Let the lattice have $N_{\text{cell}}$ unit cells, $q$ degrees of freedom per cell ($n = qN_{\text{cell}}$ in total), and $s$ species of reference operator. For each species build $M_s$ with

one *column per translate* and one *row per degree of freedom*, entry 1 where that translate has support — literally "roll the operator and write down what it touches". Then

$$\log_p \mathrm{GSD} \;=\; n - \sum_s \operatorname{rank} M_s \;=\; (q-s)\,N_{\mathrm{cell}} \;+\; \sum_s \dim\ker M_s^{\top}, \tag{8}$$

where $\dim\ker M_s^{\top}$ counts the *relations* of that species (products of translates equal to the identity).

**Why "$\log \mathrm{GSD} = \dim\ker$" looks universal but is not**

When $q = s$ the first term of (8) vanishes and the degeneracy is exactly the total number of relations. The classical stencil ($q = s = 1$, giving (2)) and the toric code ($q = s = 2$) both sit in that corner — which is precisely why the slogan appears to hold in both and is easy to over-generalise. Off that corner the term $(q-s)N_{\mathrm{cell}}$ is present and extensive.

**The toric code, done directly.** Index qubits as (cell, direction) and write the two reference operators as bare offset lists $(\Delta n_1, \Delta n_2, \mathrm{dir})$:

```
STAR      = [(0,0,h), (-1,0,h), (0,0,v), (0,-1,v)]
PLAQUETTE = [(0,0,h), (0,1,h),  (0,0,v), (1,0,v)]
```

Roll each over all $L_1 L_2$ cells, one column per translate, and eliminate:

| $(L_1, L_2)$ | (2,2) | (3,3) | (4,5) | (5,7) | (6,4) | (8,8) | (9,9) | (12,7) |
|---|---|---|---|---|---|---|---|---|
| $n$ | 8 | 18 | 40 | 70 | 48 | 128 | 162 | 168 |
| rank (each species) | 3 | 8 | 19 | 34 | 23 | 63 | 80 | 83 |
| relations (each) | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| $k$ | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |

$k = 2L_1L_2 - 2(L_1L_2 - 1) = 2$, so GSD $= 4$ on every torus — the result of §10, obtained with nothing but translation and Gaussian elimination. The cyclic-code argument of §10.1 gets the same thing for all $L$ at once instead of size by size; that, and not the count itself, is what the algebra buys.

**The trap: do not stack the species**

It is tempting to stack all species into one square support matrix and take its kernel. That is wrong. Stacking the toric code's stars and plaquettes into a single $2L_1L_2 \times 2L_1L_2$ binary matrix and computing $\dim\ker$ gives

$$(2,2) \to 4, \quad (3,3) \to 6, \quad (4,4) \to 8, \quad (5,7) \to 2, \quad (6,4) \to 4,$$

not 2. A binary support vector records *which qubit*, not *which Pauli*, so stacking lets an $X$ cancel against a $Z$ on the same qubit — which does not happen. The $X$ and $Z$ sectors must stay in separate blocks (the CSS structure) and their ranks be summed. A purely classical model has one sector, so the issue never arises there — another way the single-species case is deceptively simple.

**Stars only.** Dropping the plaquettes gives $q = 2$, $s = 1$, and the first term survives: $k = L_1L_2 + 1$, verified at $(2,2), (3,3), (4,5), (5,7), (6,4)$ as $k = 5, 10, 21, 36, 25$.

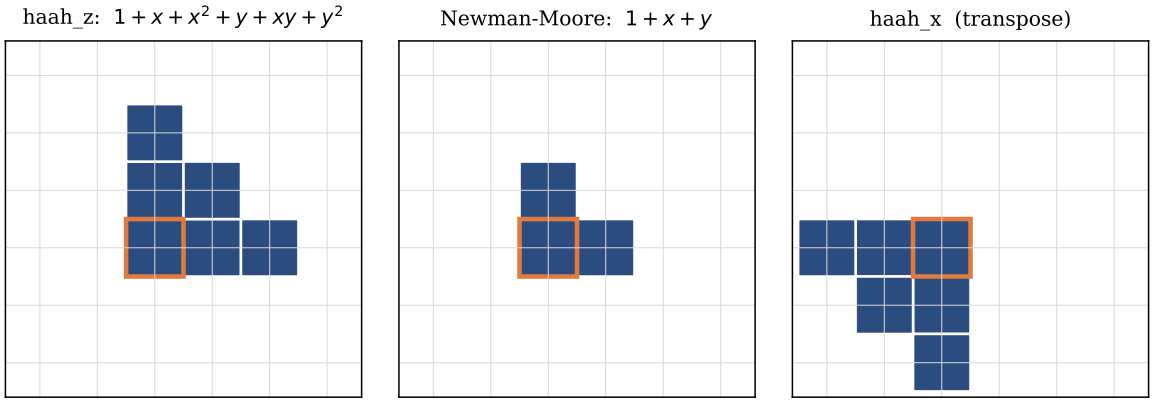



**Figure 1:** Three stencils. Orange outline marks the origin. The Haah-type corner rule (left) mixes $x$ and $y$ irreducibly; `haah_x` (right) is its reversal $\vec{s} \mapsto -\vec{s}$, i.e. the transpose operator, and therefore has the same $N_c$ for every lattice.

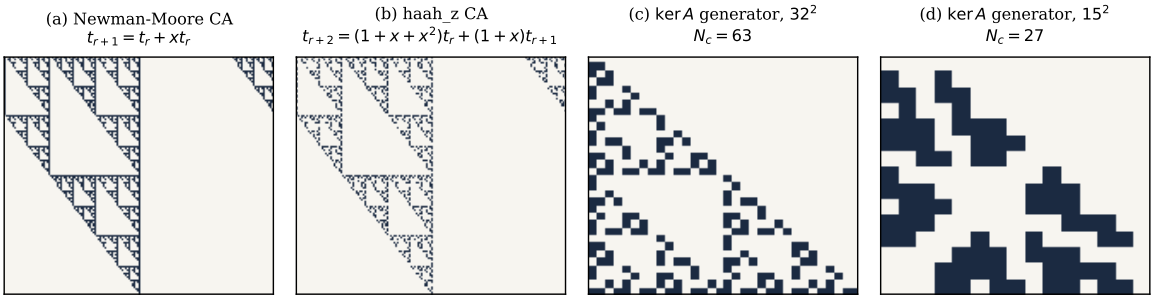



**Figure 2:** (a,b) The two stencils read as linear cellular automata from a point seed: both generate self-similar, Sierpiński-type light cones. (c,d) Actual kernel basis vectors of $A$ on the $32 \times 32$ and $15 \times 15$ tori for `haah_z`, computed by Gaussian elimination over GF(2). These are the model's fractal subsystem symmetries: rigid, non-deformable, supported on a fractal.

## 12 Fractal subsystem symmetries and UV/IR mixing

Now break the product structure. If the stencil genuinely mixes $x$ and $y$ — terms like $xy$ or $xy^{-1}$ that do not factor into an $x$-piece times a $y$-piece — the tensor argument fails, and the behaviour changes qualitatively.

**The symmetries are fractals.** Solve (3) for the topmost row: since $f = 1 + x + x^2 + y + xy + y^2$, the constraint reads

$$t_{r+2} \;=\; (1 + x + x^2)\, t_r \;+\; (1 + x)\, t_{r+1}, \tag{9}$$

a two-step linear cellular automaton over $\mathbb{F}_2$ acting on rows. Seeding two rows and iterating produces a self-similar pattern (Figure 2b), exactly as the Newman–Moore rule $t_{r+1} = (1+x)t_r$ produces Pascal's triangle mod 2, i.e. the Sierpiński gasket (Figure 2a). On the torus the CA must close up on itself, and the closed orbits are precisely the kernel vectors (Figures 2c,d).

**What "subsystem symmetry" means**

An ordinary global symmetry acts on *everything* (one generator). A *subsystem* symmetry acts only on a rigid subset — a line, a plane, or here a fractal — and there is one generator per such subset, so the number of generators grows with system size. The generators are not deformable: unlike a Wilson loop in a topological phase, you cannot smoothly move a fractal generator, which is why the associated excitations have restricted mobility (fractons). $N_c$ is exactly the number of independent generators.

**Consequences of the growth.** Two features follow immediately and are the standard diagnostics for the non-liquid branch of the classification map:

- **Sub-extensive but unbounded GSD.** For the toric code GSD $= 4$ regardless of $L$; here $N_c$ grows, so $\log_2 \mathrm{GSD} \to \infty$ with system size, while remaining $O(L)$ rather than $O(L^2)$ (Theorem 13.1 gives $N_c \le 2\min(L_x, L_y)$).
- **UV/IR mixing.** $N_c$ depends on *arithmetic* properties of $L_x, L_y$, not on the topology of the torus. A microscopic change — one extra row — can change the degeneracy by orders of magnitude. Concretely (§15), $N_c(256, 256) = 511$ but $N_c(256, 255) = 2$. No local, coarse-grained field theory can reproduce that; the ultraviolet detail (parity of $L$) controls an infrared observable.

**The diagnostic worth remembering**

Checking whether $a(x, y)$ keeps its product structure after adding a new coupling is not a technical step — it is the first diagnostic for whether the resulting phase is still ordinary topological order or something richer. If the polynomial factors direction by direction, expect a constant GSD; if it mixes, expect fractal symmetries and number-theoretic size dependence.

For a genuinely mixed $a(x, y)$ there are two routes: numerically, build the finite convolution matrix and compute its null space (what `nc_sweep.py` does); analytically, work with the ideal generated by the components and compute

$$\dim \ker = \deg \gcd\big(a_1, \dots, a_k,\ x^{L_1} - 1,\ y^{L_2} - 1\big),$$

eliminating one variable at a time by resultants — Haah's method for 3D fracton codes. Part III carries that programme out completely for (3).

# Part III — The closed form

## 13 Statement

**Theorem 13.1** (Closed form for the `haah_z` stencil at $p = 2$)**.** *Let $f = 1 + x + x^2 + y + xy + y^2$ over $\mathbb{F}_2$, let $A$ be multiplication by $f$ on $\mathcal{R}_L = \mathbb{F}_2[x, y]/(x^{L_x} - 1, y^{L_y} - 1)$, and write the 2-adic factorisations*

$$L_x = 2^a m_x, \qquad L_y = 2^b m_y, \qquad m_x, m_y \text{ odd.}$$

*Let $\omega \in \mathbb{F}_4$ satisfy $\omega^2 + \omega + 1 = 0$ and define*

$$N(m_x, m_y) \;=\; \deg \gcd_{\mathbb{F}_4[y]}\Big( y^{m_y} + 1 \;,\; (\omega y + \omega^2)^{m_x} + 1 \Big). \tag{10}$$

*Then*

$$\boxed{N_c(L_x, L_y) \;=\; 2\,\min\!\big(2^a, 2^b\big)\,N(m_x, m_y) \;-\; \delta_{a,b}} \tag{11}$$

*with* $\delta_{a,b} = 1$ *if* $a = b$ *(that is,* $v_2(L_x) = v_2(L_y)$*) and* $0$ *otherwise.*

Three things deserve emphasis.

- **The two side lengths decouple completely.** The power-of-two parts enter only through $\min(2^a, 2^b)$ and the flag $\delta_{a,b}$; the odd parts enter only through $N(m_x, m_y)$.
- $N$ **is an honest closed form**, not a disguised rank computation: it is the degree of a gcd of two explicit polynomials over $\mathbb{F}_4$, computable in $O(m_y \log m_x)$ field operations rather than the $O((L_x L_y)^3)$ Gaussian elimination of `nc_sweep.py`. Also $N \geq 1$ always, so $N_c \geq 1$: the ground space is never trivial.
- $N$ **is genuinely number-theoretic.** It is a cyclotomic point count (§16) and there is no elementary formula for it in terms of $m_x, m_y$ alone. That irreducible arithmetic core *is* the UV/IR sensitivity. Everything else about the formula is elementary.

# 14 Derivation

## 14.1 Splitting the ring

Over $\mathbb{F}_2$, characteristic-2 Frobenius gives $x^{L_x} - 1 = (x^{m_x} - 1)^{2^a}$ with $x^{m_x} - 1$ squarefree (since $m_x$ is odd). Hensel lifting therefore splits $\mathcal{R}_L$ into local factors indexed by pairs of roots of unity $(\zeta, \eta)$ with $\zeta^{m_x} = \eta^{m_y} = 1$, each of the form

$$F[s,t]/(s^A, t^B), \qquad A = 2^a, \quad B = 2^b, \quad s = x - \zeta, \quad t = y - \eta,$$

with $F$ the residue field of that factor. This is the GF(2) analogue of the Brillouin zone decomposition of §9: the pairs $(\zeta, \eta)$ are the momenta, and the nilpotents $s, t$ record the inseparable (2-adic) part.

Multiplication by $f$ is invertible on a factor unless $f(\zeta, \eta) = 0$, so only points of the curve $f = 0$ contribute:

$$N_c \;=\; \sum_{\substack{\zeta^{m_x} = \eta^{m_y} = 1 \\ f(\zeta,\eta) = 0}} \kappa(\zeta, \eta), \tag{12}$$

with $\kappa$ a *local multiplicity* depending only on $(A, B)$ and the local shape of $f$. Since $f$ is quadratic its local expansion is exact:

$$f(\zeta + s,\ \eta + t) \;=\; f(\zeta, \eta) \;+\; (1 + \eta)s \;+\; (1 + \zeta)t \;+\; (s^2 + st + t^2). \tag{13}$$

## 14.2 The curve has exactly one singular point

$f(1, y) = 1 + y^2 = (1 + y)^2$, so the only solution with $\zeta = 1$ is $(1, 1)$, and symmetrically for $\eta = 1$. Therefore:

- $(\zeta, \eta) = (1, 1)$: both linear coefficients in (13) vanish and $f = s^2 + st + t^2$ — a *nodal* (singular) point, present for every $L_x, L_y$.
- Every other point is *smooth*, with both linear coefficients non-zero.

### 14.3 Local multiplicities

**Lemma 14.1** (Smooth point). *If* $f_{\mathrm{loc}} = \alpha s+\beta t+(s^2+st+t^2)$ *with* $\alpha\beta \neq 0$, *then* $\dim\ker(f_{\mathrm{loc}}\cdot) = \min(A,B)$ *on* $F[s,t]/(s^A,t^B)$.

*Proof.* $\partial f/\partial t = \beta$ is a unit, so in $F[[s,t]]$ the Weierstrass preparation theorem gives $f = (\text{unit})\cdot(t-\varphi(s))$ with $\varphi(0)=0$. If $B \geq A$ then $\varphi(s)^B \equiv 0 \bmod s^A$ (characteristic 2, $B = 2^b$), so $t \mapsto t+\varphi(s)$ is an automorphism of $F[s,t]/(s^A,t^B)$ and $\dim\ker(f\cdot) = \dim \mathrm{Ann}(t) = A = \min(A,B)$. Swap the roles of $s,t$ if $A > B$. □

**Lemma 14.2** (Nodal point). *If* $f_{\mathrm{loc}} = s^2+st+t^2$ *then* $\dim\ker(f_{\mathrm{loc}}\cdot) = 2\min(A,B) - \delta_{A,B}$.

This follows from the $\mathbb{F}_4$-factorisation $s^2+st+t^2 = (s+\omega t)(s+\omega^2 t)$ into two "smooth"-type factors, and was verified directly by explicit kernel computation for all $A,B \in \{1,2,4,8,16\}$:

$D(A,B) =$

| $A\backslash B$ | 1 | 2 | 4 | 8 | 16 |
|---|---|---|---|---|---|
| 1 | 1 | 2 | 2 | 2 | 2 |
| 2 | 2 | 3 | 4 | 4 | 4 |
| 4 | 2 | 4 | 7 | 8 | 8 |
| 8 | 2 | 4 | 8 | 15 | 16 |
| 16 | 2 | 4 | 8 | 16 | 31 |

Every entry equals $2\min(A,B) - \delta_{A,B}$.

Writing $P$ for the total number of points $(\zeta,\eta)$ on the curve, (12) becomes

$$N_c = (P-1)\min(A,B) + \big(2\min(A,B) - \delta_{A,B}\big) = \min(A,B)\,(P+1) - \delta_{A,B}. \tag{14}$$

### 14.4 The conic factors over $\mathbb{F}_4$: the fractal stencil is two straight lines

Homogenise: $Q = x^2+y^2+z^2+xy+yz+zx$. Its polar form has radical $(1:1:1)$, and $Q(1,1,1) = 0$, so the conic is *degenerate* — a pair of conjugate lines meeting at $(1:1:1)$. Explicitly,

$$\boxed{f(x,y) \;=\; \big(x+\omega y+\omega^2\big)\big(x+\omega^2 y+\omega\big) \quad \text{over } \mathbb{F}_4.} \tag{15}$$

**The punchline of Part III**

The "fractal" Haah stencil is, algebraically, **two straight lines over** $\mathbb{F}_4$. Everything erratic about $N_c$ comes from asking how many roots of unity happen to lie on a line.

Frobenius $u \mapsto u^2$ swaps the two branches and preserves $\mu_{m_x} \times \mu_{m_y}$, and the branches meet only at $(1,1)$. Hence

$$P(m_x,m_y) = 2N(m_x,m_y) - 1, \qquad N = \#\big\{\eta \in \mu_{m_y} \;:\; \omega\eta+\omega^2 \in \mu_{m_x}\big\}. \tag{16}$$

Because $y^{m_y}-1$ is squarefree, that count is exactly the gcd degree (10). Substituting $P+1 = 2N$ into (14) gives (11). ■

## 15 Explicit special cases

Let $\mu = \min(2^a, 2^b)$.

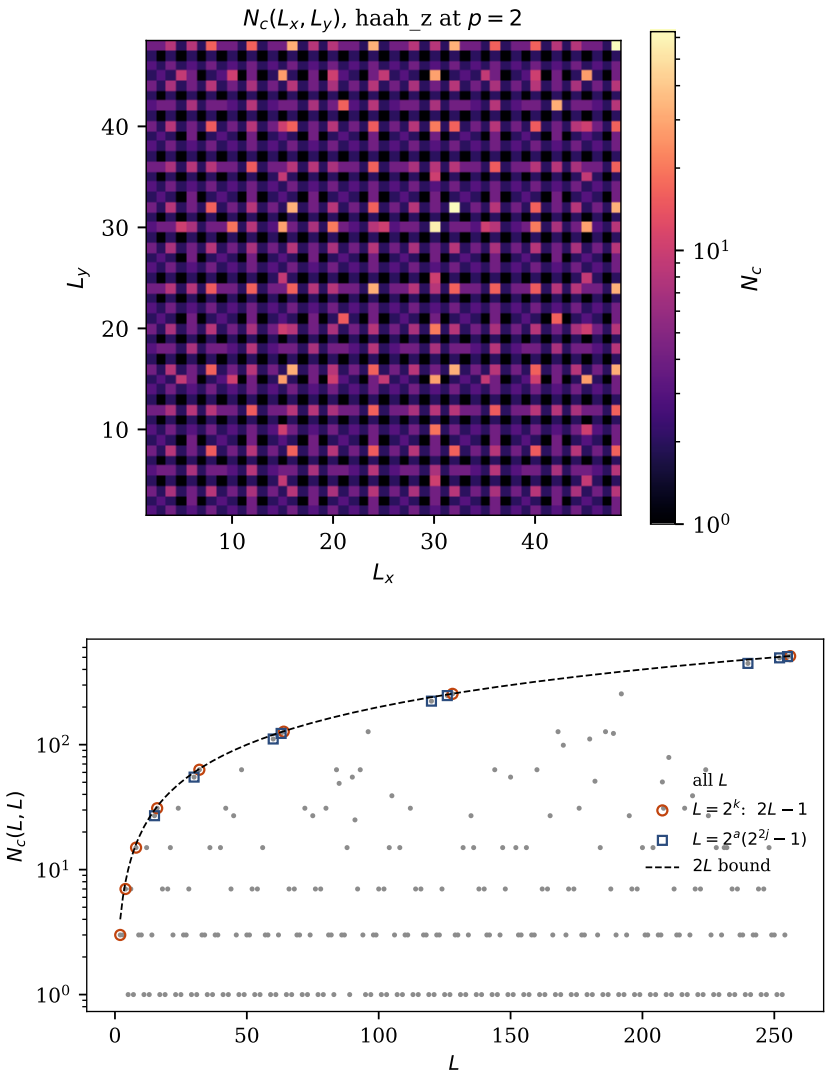



**Figure 3:** Top: $N_c(L_x, L_y)$ from (11) on the full $2 \le L_x, L_y \le 48$ grid (log colour scale); the bright cells are the $v_2(L_x) = v_2(L_y)$ diagonal families. Bottom: $N_c(L, L)$ against $L$, with the two extremal families marked and the bound $N_c \le 2L$ shown dashed. The scatter between them is the number-theoretic term $N$.

| regime | $N_c$ | GSD |
|---|---|---|
| generic ($N = 1$, see §16) | $2\mu - \delta_{a,b}$ | $2^{2\mu - \delta_{a,b}}$ |
| $L_x, L_y$ both odd, generic | 1 | 2 |
| $L_x = L_y = 2^k$ | $2^{k+1} - 1 = 2L - 1$ | $2^{2L-1}$ |
| $L_x = 2^a,\ L_y = 2^b,\ a \neq b$ | $2^{\min(a,b)+1}$ | $2^{2\min(L_x, L_y)}$ |
| $L_x = L_y = L = 2^a m$ | $2^{a+1} N(m, m) - 1$ | — |
| $L_x = L_y = 2^a(2^k - 1)$, $k$ *even* | $2^a(2^{k+1} - 4) - 1$ | — |
| $L_x = L_y = 2^a(2^k - 1)$, $k$ *odd* | $2^{a+1} - 1$ | — |

The $k$-even row is the maximal-degeneracy family: $\mu_{2^k - 1} = \mathbb{F}_{2^k}^{\times}$ and $3 \mid 2^k - 1$, so the branch condition becomes $u + v = 1$ with $u, v \in \mathbb{F}_{2^k}^{\times}$, giving $N = 2^k - 2$. It reproduces

$$N_c(15, 15) = 27, \quad N_c(63, 63) = 123, \quad N_c(255, 255) = 507, \quad N_c(1023, 1023) = 2043, \quad N_c(4095, 4095) = 8187$$

i.e. $N_c \approx 2L$ — the same linear growth as the pure powers of two, and the largest possible.

Two consequences worth stating separately:

- $N_c \le 2\min(L_x, L_y)$ always, so GSD $\le 2^{2\min(L_x,L_y)}$ — sub-extensive, as expected for a fracton-like code.
- A single change of parity destroys the degeneracy: $N_c(2^k, 2^k) = 2^{k+1} - 1$, but $N_c(2^k, 2^k \pm 1)$ collapses to 2. For example $N_c(256, 256) = 511$ while $N_c(256, 255) = 2$.

## 16 When is $N > 1$? The number-theoretic part

$N = 1$ means the only curve point on the torus $\mu_{m_x} \times \mu_{m_y}$ is the singular one, $(1, 1)$. This is the *generic* situation.

**Sufficient criterion for $N = 1$.** If $d = \mathrm{lcm}\big(\mathrm{ord}_{m_x}(2), \mathrm{ord}_{m_y}(2)\big)$ is *odd*, then $\mathbb{F}_4 \not\subset \mathbb{F}_{2^d}$; writing $\xi + \omega\eta + \omega^2 = 0$ in the basis $\{1, \omega\}$ of $\mathbb{F}_{2^{2d}}$ over $\mathbb{F}_{2^d}$ forces $\xi = \eta = 1$. Hence $N = 1$ and $N_c = 2\mu - \delta_{a,b}$. (E.g. any $m \mid 2^k - 1$ with $k$ odd: $m = 7, 31, 127, 73, \dots$)

**Sufficient criterion for $N > 1$.** If $3 \mid \gcd(m_x, m_y)$ then $\omega \in \mu_{m_x} \cap \mu_{m_y}$ and $(\xi, \eta) = (\omega, \omega^2)$ lies on a branch, so $N \ge 2$ and $P \ge 3$. This is why every $L$ divisible by 3 shows enhanced degeneracy in the sweep data.

**Neither criterion is necessary.** The set $\{m : N(m, m) > 1\}$ is the set of multiples of the *levels* $\ell = \mathrm{lcm}\big(\mathrm{ord}(\eta), \mathrm{ord}(\omega\eta + \omega^2)\big)$ of the non-singular points. The smallest levels are

$$\ell = 3, \quad 85 = 5 \cdot 17, \quad 91 = 7 \cdot 13, \quad 341 = 11 \cdot 31, \quad 1387 = 19 \cdot 73, \ \dots$$

so for $m$ coprime to 3 the first $m$ with $N(m, m) > 1$ are 85 (giving $N_c = 49$) and 91 (giving $N_c = 25$). Both were checked directly against brute force ($\approx 1.3\,\mathrm{s}$ each); the next level, $m = 341$, is not reachable that way (116281 sites, 1.7 GB bit-packed), and the formula gives $N_c(341, 341) = 241$.

**Size of $N$.** With $\mu_{m_x}, \mu_{m_y} \subset \mathbb{F}_q$, $q = 2^d$, expanding the coset indicators in multiplicative characters writes $N$ as an average of Jacobi sums,

$$N = \frac{1}{e_x e_y} \sum_{\chi,\psi} J(\chi, \psi), \qquad e_x = \frac{q-1}{m_x}, \quad e_y = \frac{q-1}{m_y},$$

so by Weil's bound

$$N = \frac{m_x m_y}{q} + O(\sqrt{q}). \tag{17}$$

This is the closed form in the analytic sense, and it explains the erratic look of the data: $N_c$ tracks $m_x m_y / 2^{\mathrm{ord}}$, which jumps whenever the multiplicative order of 2 changes. **This is where the UV/IR mixing lives**: the multiplicative order of 2 modulo $L$ is not a smooth function of $L$, and no coarse-graining can make it one.

## 17 The method generalises: Newman–Moore in closed form

To show that §14.1–§14.4 is a method and not a one-off, apply it unchanged to the Newman–Moore rule $f_{\mathrm{NM}} = 1 + x + y$ (the Sierpiński stencil of Figure 1). It is *easier*, because the curve has no singular point: $\partial f/\partial x = \partial f/\partial y = 1$ never vanishes, so every zero is smooth and contributes $\kappa = \min(A, B)$ by Lemma 14.1. Hence

$$N_c(L_x, L_y) = \min\big(2^a, 2^b\big)\, N_{\mathrm{NM}}(m_x, m_y), \qquad N_{\mathrm{NM}} = \deg \gcd_{\mathbb{F}_2[y]}\Big(y^{m_y} + 1,\ (1+y)^{m_x} + 1\Big), \tag{18}$$

equivalently $N_{\mathrm{NM}} = \#\{\zeta \in \mu_{m_x} : 1 + \zeta \in \mu_{m_y}\}$. Note the **absence of a $\delta_{a,b}$ term**: in (11) that correction came entirely from the nodal point, which this stencil does not have. Checked against brute force on the full grid $2 \le L_x, L_y \le 24$: **529/529 agree**.

| $m$ | 1 | 3 | 5 | 7 | 9 | 11 | 13 | 15 | 17 | 21 | 27 | 31 | 63 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $N_{\mathrm{NM}}(m,m)$ | 0 | 2 | 0 | 6 | 2 | 0 | 0 | 14 | 0 | 8 | 2 | 30 | 62 |

The contrast with (11) is entirely about the constant term: $f_{\mathrm{NM}}(1,1) = 3 \equiv 1$ over $\mathbb{F}_2$, so $(1,1)$ is *not* on the curve, and on a $2^k \times 2^k$ torus there are no other candidate points at all — $N_c = 0$, a unique ground state. That is the Frobenius dichotomy (stencil weight $w = f(1,1) \not\equiv 0$) seen as the $m_x = m_y = 1$ corner of (18). Over $\mathbb{F}_3$ the weight $w = 3 \equiv 0$ instead, $(1,1)$ is on the curve, and the dichotomy flips: GSD $= 3^L$ on every $3^k \times 3^k$ torus (verified at $L = 3, 9, 27$, giving $N_c = L$). The same stencil is maximally rigid over one field and maximally degenerate over another.

| | $f_{\mathrm{NM}} = 1 + x + y$ | `haah_z` |
|---|---|---|
| $f(1,1)$ over $\mathbb{F}_2$ | 1 — not a zero | 0 — a nodal zero |
| singular points | none | one, at $(1,1)$, for every $L$ |
| $N_c$ on $2^k \times 2^k$ | 0 | $2L - 1$ |
| $\delta_{a,b}$ correction | absent | present |
| can $N_c$ vanish? | yes, generically | never ($N_c \ge 1$) |

## 18 Numerical validation

Every claim above was re-derived from scratch for this document by an independent implementation (brute-force circulant rank over GF(2); separate $\mathbb{F}_4$ polynomial arithmetic for the gcd; explicit local kernels for $\kappa$).

| check | result |
|---|---|
| Full grid $2 \le L_x, L_y \le 40$: brute-force $\dim\ker A$ vs. (11) | 1521/1521 agree |
| Nodal multiplicity $D(A,B)$ by explicit kernel, $A, B \in \{1,2,4,8,16\}$, vs. $2\min(A,B) - \delta_{A,B}$ | 25/25 agree |
| Smooth multiplicity by explicit kernel vs. $\min(A,B)$, same range | 25/25 agree |
| $\mathbb{F}_4$ factorisation (15) expanded symbolically vs. $f$ | exact |
| Special cases $N_c(15,15), N_c(63,63), N_c(255,255), N_c(1023,1023), N_c(4095,4095)$ | 27, 129, 507, 2043, 8187 |
| $N_c(85,85) = 49$, $N_c(91,91) = 25$, $N_c(256,256) = 511$, $N_c(256,255) = 2$, $N_c(64,64) = 127$ | all agree |
| $P(m,m) = 2N(m,m) - 1$ for the 21 tabulated odd $m$ up to 105 | 21/21 agree |
| Levels: odd $m < 700$ coprime to 3 with $N(m,m) > 1$ vs. multiples of $85, 91, 341, 1387$ | exact match |
| Kernel basis dimension by column reduction at $32^2$ and $15^2$ vs. formula | 63, 27 |
| Toric code by direct rolling (no polynomials), 10 lattices incl. coprime and rectangular | GSD = 4 throughout |
| General identity (8) vs. direct rank, toric code and stars-only | all agree |
| Newman–Moore (18) vs. brute force, grid $2 \le L_x, L_y \le 24$ | 529/529 agree |
| Newman–Moore at $p = 3$, $L = 3, 9, 27$ vs. $N_c = L$ | all agree |

The odd-part input $P(m,m) = 2N(m,m) - 1$:

| $m$ | 1 | 3 | 5 | 7 | 9 | 11 | 13 | 15 | 17 | 19 | 21 | 23 | 25 | 27 | 33 | 45 | 63 | 85 | 91 | 93 | 105 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $P$ | 1 | 3 | 1 | 1 | 3 | 1 | 1 | 27 | 1 | 1 | 15 | 1 | 1 | 3 | 3 | 27 | 123 | 49 | 25 | 63 | 39 |

(all odd $m \le 31$ not listed have $P = 1$).

# Part IV — Reading the code

## 19 `nc_sweep.py`, line by line

The script is short and does exactly what Part I describes. Worth noting, in order:

```
HAAH_Z_SHIFTS = [(0,0),(0,1),(0,2),(1,0),(1,1),(2,0)]
```

with the comment "shifts are `(d_row, d_col) = (e2, e1)`". Translating by §4, $x \leftrightarrow$ column shift and $y \leftrightarrow$ row shift, this is $f = 1 + x + x^2 + y + xy + y^2$. Any change to this list changes the polynomial and therefore the entire analysis — it is the only physics input.

```
r, c = np.divmod(np.arange(N), Lx)
rows = ((r + d2) % Ly) * Lx + (c + d1) % Lx
A[rows, np.arange(N)] += coeff
```

Column $k$ is the image of the delta function at site $k$, so $A$ is the matrix of "multiply by $f$" in the basis $\{x^a y^b\}$. The double `%` is where the group ring's $x^{L_x} = 1$, $y^{L_y} = 1$ enters. Note $(At)_{\vec{r}} = \sum_{\vec{s}} t_{\vec{r}-\vec{s}}$ with this convention — the *correlation*, i.e. multiplication by the reciprocal polynomial — which is why `haah_x` is defined as the negated shift list and why the two give identical $N_c$: they are transposes, and $\operatorname{rank} A = \operatorname{rank} A^\top$. (This also means the sweep never needs to be run twice.)

**`rank_gf2` is the reason large lattices are reachable.** Rows are bit-packed into `uint64` words, so an entire row XOR is $\lceil N/64 \rceil$ word operations instead of $N$ byte operations — roughly a 64× constant-factor win, and 8× less memory than `uint8`. The pivot loop is textbook Gauss–Jordan with XOR replacing subtraction (in characteristic 2 they are the same). Two small remarks:

- `np.pad(..., (-M.shape[1]) % 8).view(np.uint64)` pads to a multiple of 8 bytes so the `uint64` view is legal — a necessary detail, easy to break.
- Complexity is $O(N^3/64)$ with $N = L_x L_y$, i.e. $O((L_x L_y)^3)$. At $L = 128$ that is $N = 16384$ and $\sim 7 \times 10^{10}$ word ops; the closed form (11) does the same job in $O(m_y \log m_x)$ $\mathbb{F}_4$ operations. Concretely: $85 \times 85$ and $91 \times 91$ still take $\approx 1.3\,\mathrm{s}$ each by brute force, but $341 \times 341$ needs 1.7 GB even bit-packed and $\sim 10^{13}$ word operations, while the gcd answers in microseconds. Cost is not the obstacle at small $L$; it is the obstacle exactly at the sizes where the answer stops being generic (§16).

**`rank_gfp` handles $p > 2$ generically** , using `pow(a, p-2, p)` for the inverse (Fermat's little theorem — correct only because $p$ is prime, which the CLI does not check). Note that Theorem 13.1 is **specific to** $p = 2$: the split-conic argument (15) is a characteristic-2 statement, and `-p 3` changes both the base field and how $f$ factors.

**The driver's defaults are subtle.** With a single `-lx` value, `-ly` defaults to $2..L_x$; with a range, to the same range, and without `-rect` only the diagonal $L_x = L_y$ is swept. Output is flushed per row, so a long sweep is resumable by inspection. The default `-lx 2-128` on the diagonal is already at the edge of feasibility for the reason above.

**One suggestion.** `Nc()` rebuilds the full $N \times N$ dense matrix for every lattice; for $p = 2$ the matrix is determined by one row, so a rank computation that never materialises $A$ (or a direct call to the closed form, cross-checked against brute force on small sizes as in §18) removes the memory wall entirely. The CSV schema `Lx,Ly,Nc` is already the right one for that swap.

**What the code is really computing**

`nc_sweep.py` counts, for each torus, **the number of independent fractal subsystem symmetries** of the stencil Hamiltonian — equivalently the dimension of its ground-state code, equivalently $\dim \mathrm{Ann}_{\mathcal{R}_L}(f)$. The kernel *basis* it can also return is a list of the symmetry generators themselves, and those are the pictures in Figure 2.

## 20 Scope and open directions

- The derivation is specific to `haah_z`/`haah_x` at $p = 2$, but §17 carries the same three steps through for Newman–Moore, and §11 shows the underlying count needs no polynomial at all.
- The *method* is general for any single-species translation-invariant stencil: (i) write the stencil as $f$; (ii) find the points of $f = 0$ on $\mu_{m_x} \times \mu_{m_y}$; (iii) attach the local multiplicity of each point in $F[s,t]/(s^{p^a}, t^{p^b})$. The $p$-adic/prime-to-$p$ split is a consequence of $x^{p^a m} - 1 = (x^m - 1)^{p^a}$ and holds for any $f$.
- With several operator species stacked, $A$ becomes a matrix of polynomials and the kernel becomes a *syzygy module* — harder symbolically, no harder numerically.
- The natural next targets: the same programme for $\Delta_6$ (whose closed form $N_c = L_x + L_y + \gcd(L_x, L_y) - 2$ at $p = 2$ already hints at the cyclotomic decomposition), and the leading-form conjecture $N_c(L = p^k) = dL - (r - 1)$ recorded in the vault note on the Frobenius dichotomy, which the local-multiplicity calculus of §14.3 looks well-suited to prove.

## A Glossary

*group* — A set with one associative operation, an identity, and inverses. Here: lattice positions under translation, $\mathbb{Z}_{L_x} \times \mathbb{Z}_{L_y}$.

*ring* — A set with $+$ and $\times$ obeying the usual rules. Here: $\mathbb{F}_2$, and polynomials over it.

*field* — A ring where every non-zero element is invertible. Here $\mathbb{F}_2$, $\mathbb{F}_4$, $\mathbb{F}_{2^d}$.

*quotient ring* — $R/(m)$: $R$ with $m$ declared to be 0. Here: how $x^L = 1$ (periodicity) is imposed.

*group ring* — $\mathbb{F}_2[G]$: formal sums over a group. For $G$ a translation group it is the ring of periodic patterns, with convolution as multiplication.

*annihilator* — $\mathrm{Ann}(f) = \{t : ft = 0\}$. Here: the ground states / symmetries, whose dimension is $N_c$.

*circulant* — A matrix each of whose columns is a cyclic shift of the previous. The matrix of "multiply by $f$".

*Frobenius* — $u \mapsto u^p$; a ring homomorphism in characteristic $p$. Source of both $x^{p^a m} - 1 = (x^m - 1)^{p^a}$ and the $\omega \leftrightarrow \omega^2$ branch swap.

$\mu_m$ The group of $m$-th roots of unity in $\overline{\mathbb{F}_2}$.

$v_2(L)$ The 2-adic valuation: the exponent $a$ in $L = 2^a m$, $m$ odd.

*subsystem symmetry*
A symmetry acting on a rigid sub-manifold (line, plane, fractal) rather than the whole system; the number of generators grows with system size.

*UV/IR mixing*
Dependence of an infrared observable (here GSD) on microscopic details (here the arithmetic of $L$).

# B Reproducing the numbers

The brute force is (1) plus rank–nullity; the closed form is (11). In pseudocode:

```
def Nc_closed(Lx, Ly):
    a, mx = v2(Lx); b, my = v2(Ly)          # Lx = 2^a * mx, mx odd
    N = deg_gcd_F4(y**my + 1, (w*y + w**2)**mx + 1)
    return 2 * min(2**a, 2**b) * N - (1 if a == b else 0)
```

with $\mathbb{F}_4$ elements stored as two bits ($a + b\omega$), addition = XOR, and multiplication reduced by $\omega^2 = \omega + 1$. The gcd is the ordinary Euclidean algorithm on $\mathbb{F}_4[y]$; $(\omega y + \omega^2)^{m_x}$ is computed by square-and-multiply modulo $y^{m_y} + 1$, which is what makes the cost $O(m_y \log m_x)$.